```
GNU nano 8.7.1
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>

int main(int argc,char *argv[])
{
        int mb;

        if(argc != 2)
        {
           printf("Usage: %s <memory_in_MB>\n", argv[0]);
           return 1;
        }
        mb = atoi(argv[1]);

        char *memory = malloc(mb * 1024 * 1024);

        if (memory == NULL)
        {
           printf("Memory allocation failed\n");
           return 1;
        }

        for(int i=0; i < mb * 1024 * 1024; i += 4096)
        {
            memory[i] = 1;
        }


        printf("%d MB memory allocated.\n", mb);
        printf("Program is running... Press Ctrl+C to exit.\n");

        while(1)
        {
             sleep(1);
        }
        return 0;
```

```
vboxuser@Kartik:~$ nano exp6.c
vboxuser@Kartik:~$ gcc exp6.c -o exp6
vboxuser@Kartik:~$ chmod +x exp6
vboxuser@Kartik:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.4Gi       1.5Gi        38Mi       1.7Gi       2.9Gi
Swap:             0B          0B          0B
vboxuser@Kartik:~$ ./exp6 100
100 MB memory allocated.
Program is running... Press Ctrl+C to exit.
```

```
vboxuser@Kartik:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.5Gi       1.4Gi        38Mi       1.7Gi       2.8Gi
Swap:             0B          0B          0B
vboxuser@Kartik:~$
```